REDUCEDNESS OF THE FROBENIUS TWIST, GEOMETRIC REDUCEDNESS AND EXISTENCE OF A SMOOTH OPEN ARE EQUIVALENT, AND THEN THE RELATIVE FROBENIUS IS FINITE DOMINANT OF DEGREE P TO THE DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites twelve cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Finiteness of the relative Frobenius. Stacks Project, Varieties, Lemma lemma-relative-frobenius-finite (tag 0CC9). Makes the relative Frobenius a finite morphism, so that its degree is defined.
- Equivalent characterizations of geometric reducedness. Stacks Project, Varieties, Lemmas lemma-geometrically-reduced-p (tag 0CC6), lemma-geometrically-reduced-dense-smooth-open (tag 0CCH) and lemma-generic-points-geometrically-reduced (tag 0CCE). Supply the three implications making up the equivalence.

Source reference. Original lemma 0CCE: sources/stacks-chow/tags.txt, source lines 15504–15504. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 056V: sources/stacks-chow/tags.txt, source lines 6723–6723. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 038X: sources/stacks-chow/tags.txt, source lines 4252–4252. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04KS: sources/stacks-chow/tags.txt, source lines 5983–5983. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0CCB: sources/stacks-chow/tags.txt, source lines 15501–15501. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0CCD: sources/stacks-chow/tags.txt, source lines 15503–15503. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 02NY: sources/stacks-chow/tags.txt, source lines 3497–3497. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0CCA: sources/stacks-chow/tags.txt, source lines 15500–15500. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 054L: sources/stacks-chow/tags.txt, source lines 6644–6644. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02NL: sources/stacks-chow/tags.txt, source lines 3484–3484. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02NZ: sources/stacks-chow/tags.txt, source lines 3498–3498. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field of characteristic $p > 0$. Let X be a variety over k. The following are equivalent*

(1) *$X^{(p)}$ is reduced,*
(2) *X is geometrically reduced,*
(3) *there is a nonempty open $U \subset X$ smooth over k.*

In this case $X^{(p)}$ is a variety over k and $F_{X/k} : X \to X^{(p)}$ is a finite dominant morphism of degree $p^{\dim(X)}$.

Proof. We have seen the equivalence of (1) and (2) in Lemma 0CCE. We have seen that (2) implies (3) in Lemma 056V. If (3) holds, then U is geometrically reduced (see for example Lemma 038X) and hence X is geometrically reduced by Lemma 04KS. In this way we see that (1), (2), and (3) are equivalent.

Assume (1), (2), and (3) hold. Since $F_{X/k}$ is a homeomorphism (Lemma 0CCB) we see that $X^{(p)}$ is a variety. Then $F_{X/k}$ is finite by Lemma 0CCD. It is dominant as it is surjective. To compute the degree (Morphisms, Definition 02NY) it suffices to compute the degree of $F_{U/k} : U \to U^{(p)}$ (as $F_{U/k} = F_{X/k}|_U$ by Lemma 0CCA). After shrinking U a bit we may assume there exists an étale morphism $h : U \to \mathbf{A}^n_k$, see Morphisms, Lemma 054L. Of course $n = \dim(U)$ because $\mathbf{A}^n_k \to \mathrm{Spec}(k)$ is smooth of relative dimension n, the étale morphism h is smooth of relative dimension 0, and $U \to \mathrm{Spec}(k)$ is smooth of relative dimension $\dim(U)$ and relative dimensions add up correctly (Morphisms, Lemma 02NL). Observe that h is a generically finite dominant morphism of varieties, and hence $\deg(h)$ is defined. By Lemma 0CCA we have a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{F_{X/k}} & U^{(p)} \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h^{(p)}} \\ \mathbf{A}^n_k & \xrightarrow{F_{\mathbf{A}^n_k/k}} & (\mathbf{A}^n_k)^{(p)} \end{array}$$

Since $h^{(p)}$ is a base change of h it is étale as well and it follows that $h^{(p)}$ is a generically finite dominant morphism of varieties as well. The degree of $h^{(p)}$ is the degree of the extension $k(X^{(p)})/k((\mathbf{A}^n_k)^{(p)})$ which is the same as the degree of the extension $k(X)/k(\mathbf{A}^n_k)$ because $h^{(p)}$ is the base change of h (small detail omitted). By multiplicativity of degrees (Morphisms, Lemma 02NZ) it suffices to show that

the degree of $F_{\mathbf{A}_k^n/k}$ is p^n. To see this observe that $(\mathbf{A}_k^n)^{(p)} = \mathbf{A}_k^n$ and that $F_{\mathbf{A}_k^n/k}$ is given by the map sending the coordinates to their pth powers. $\square$